Due Date: 05.01.2026 23:59

CENG 463 – Introduction to Machine Learning HW3

As an image clustering task, you will work with a pre-built image dataset, CIFAR-10, which contains images of 10 classes such as airplanes, cars, cats, dogs, and more. The goal is to explore unsupervised learning techniques to automatically group similar images, analyze the results, and visualize the clusters.

1. Dataset Overview (10 pts)

- Load the dataset using Keras(More information: Keras CIFAR-10.):

```
from keras.datasets import cifar10
(x_train, y_train), (x_test, y_test) = cifar10.load_data()
```

- Show the dataset size and class names in a table.
- Display sample images from each class.

2. Feature Extraction and PCA (30 pts)

- Convert each image from 32×32×3 into a 3072-dimensional vector.
- Scale the vectors using StandardScaler.
- Apply PCA to reduce the dimensionality from 3072 to 50.

3. Clustering Methods (30 pts)

- Perform clustering on the PCA-transformed features using three Sklearn algorithms: K-Means, Agglomerative Clustering, DBSCAN
- Observe the clusters created by each algorithm.
- Display example images for each cluster.

Hint: Use stratified sampling to maintain class proportions and avoid crashing in Colab.

4. Visualization and Analysis (30 pts)

- Use t-SNE to reduce PCA features to 2D for visualization.
- Create colored scatter plots for each clustering algorithm.
- Observe which classes are better grouped by each algorithm.

SUBMISSION

- Jupyter Notebook (.ipynb) – should include working code and comments.
- PDF Report – describe the steps, include tables, visualizations, and brief analysis.
- File name: HW3_CENG463_StudentID.zip (Notebook + PDF).

!!!!!IMPORTANT

For each task, write a short discussion in the HW3_Report.pdf file explaining what you did, how you interpreted the results, and any important observations, insights, or limitations you identified during the analysis.

Assignment Rules:

1. In this homework, no cheating is allowed. If any cheating is detected, the homework will be graded as 0, and no further discussion will be entertained.

2. You are expected to submit your homework in groups. Therefore, it will be sufficient if only one member of the group submits the homework.

3. You must **fill in the provided Jupyter Notebook file** named:
   - CENG463_HW3_GROUPID.ipynb

   Follow the steps and instructions **provided in this file carefully.**

4. You must upload a **.zip file** to MS Teams. This file must include:
   - **CENG463_HW3_GROUPID.ipynb**

     Example: CENG463_HW3_G01.ipynb

   - **CENG463_HW3_GROUPID_Report.pdf**

     Example: CENG463_HW3_G01_Report.pdf

5. The .zip file must be named in the following format: **CENG463_HW3_GROUPID.zip**
   Example: CENG463_HW3_G01.zip
6. Please be aware that if you do not follow the assignment rules regarding export format and naming conventions, you will lose points.